Group Assignment 2

Tuning Deep Neural Networks
on real-life Business Data

Background and Goal:

The goal of this assignment is to gather experience on the sensitivity of the algorithm to different kinds of tuning parameters.

As you will see, there is **no code submission** for this assignment; only a presentation is due.

The goal of this assignment is to **reward you for doing the work and, most importantly, for learning from it**.

The presentation itself carries relatively little weight toward your final grade (typically **5%; see the syllabus**). This is intentional. If your only goal were to complete the assignment, you could probably do so in an hour by relying heavily on tools such as Codex or Claude Code. But you would learn very little from doing that.

Instead, think of the assignment as preparation for **Test 2**. The test will include questions related to this assignment and may require you to **write and reason through code using pen and paper**.

So, while you are welcome to use AI tools as part of your learning process, make sure that **you understand the code, the decisions you made, and why the code works**. That understanding, not simply producing working code, is the real objective of the assignment.

Data:

All data is contained in *pricing.csv (s*ame data as assignment 1). The data is copyrighted and confidential.

One line in the data represents a product selling on the company's e-commerce website. If a product goes out-of-stock and then becomes in-stock again a new line is created in the data for that product

- sku: stock keeping unit
- price: the price of the product on the website
- quantity: total quantity sold
- order: identifies the nth time the product was in-stock and selling on the website
- duration: how long the product appeared on the site before going out-of-stock
- category: product category

Note 1: All categorical variables are integer encoded (not necessarily consecutively). All numeric variables are divided by a constant.

Variables

Input variables

- sku
- price
- order
- duration
- category

Response variable:

- quantity

Tuning parameters

Batch size, number of hidden layers, number hidden neurons, hidden activation functions (sigmoid, tanh, relu, leaky relu, prelu, elu), optimizers (plain SGD, momentum, nesterov, adagrad, rmsprop, adam, learning rate scheduling), sequence models, …, anything we have seen in class.

Deliverables:

Presentation (pdf or PowerPoint)
-Useful insights on the sensitivity to tuning parameters
-Details on training and testing
-Statement of who did what to show contributions of all team members

Deadline:

See syllabus.

Submission link:

https://www.dropbox.com/request/fnbc8zhd3ubefgqvkai3

Groups

See groups_from_Zaretzki.xlsx